

Continuous Microwave Exfoliation Reactor for Graphene Nanoplatelets Manufacture

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Exfoliated Graphene : Current Challenges

APPLICATIONS

Polymer Composites

Conductive Coatings & Inks

Fuel Cell Batteries

Catalysis

Ultracapacitors

3D Printing

Graphene

nanosheets

nanoparticles

nanoplatelets

REQUIREMENTS

1. High Quality
2. Low Cost
3. Scalability
4. Reproducibility
5. Purity

What do we offer?

New Technology

“A unique, continuous, microwave driven process to generate high quality bulk Graphene on an industrial scale”

Cost

“A low cost, high yield and scalable process”

Presentation Content

I. BTU Introduction

II. Proof of Concept

- MW Technology
- Reactant Materials
- System Description
- Thermal Profile
- XRD results
- TGA/DSC results
- Microscopy results
- Summary

III. Scale Up Model

- Flow Reactor : 60 Tons/Year capacity
- Comsol Model :Flow Reactor Design
- Flow Reactor Scaling Conditions

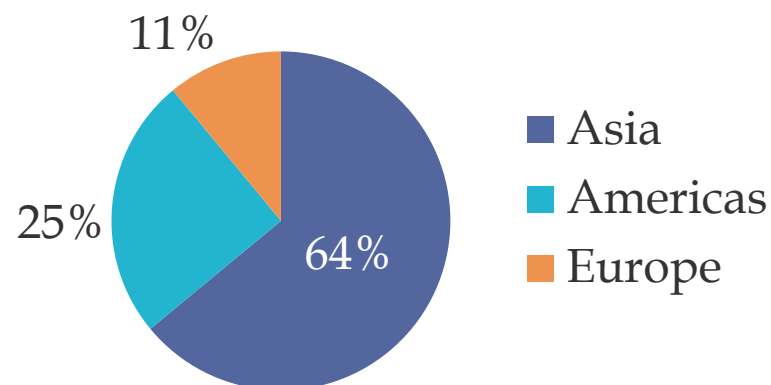
IV. Summary



I. BTU Introduction

Corporate Facts

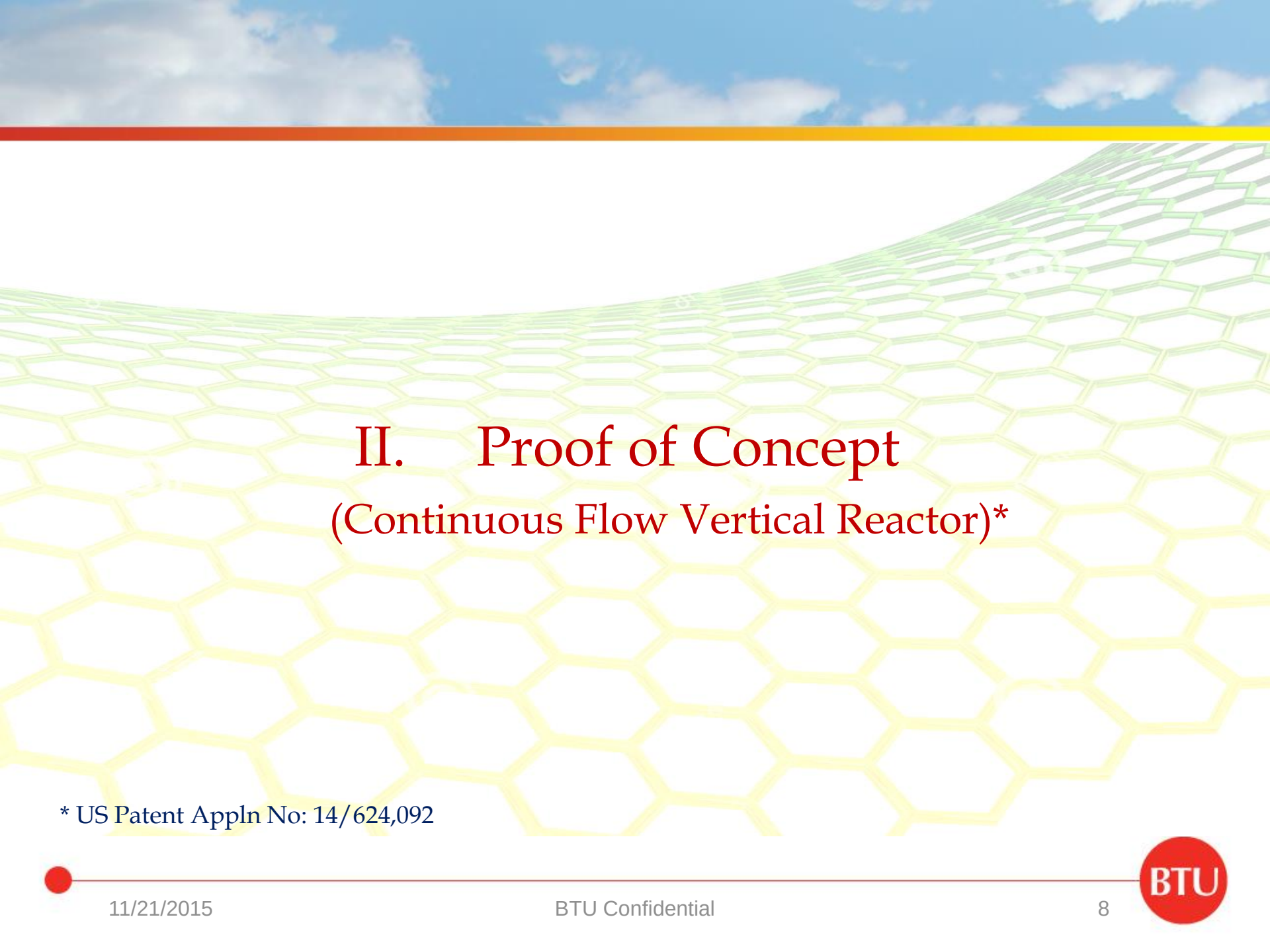
- **BTU Background**
 - Thermal process equipment for **in-line, high volume** mfg.
 - Founded in 1950
 - Acquired by AMTECH 2015
- **Headquarters in Billerica, MA**
 - Manufacturing and engineering in **US** and **China**
 - Worldwide sales and support
 - 280 employees worldwide
- **Revenue FY 2014, \$58.5 M**



2014 Shipments by Territory

Target Markets for BTU products

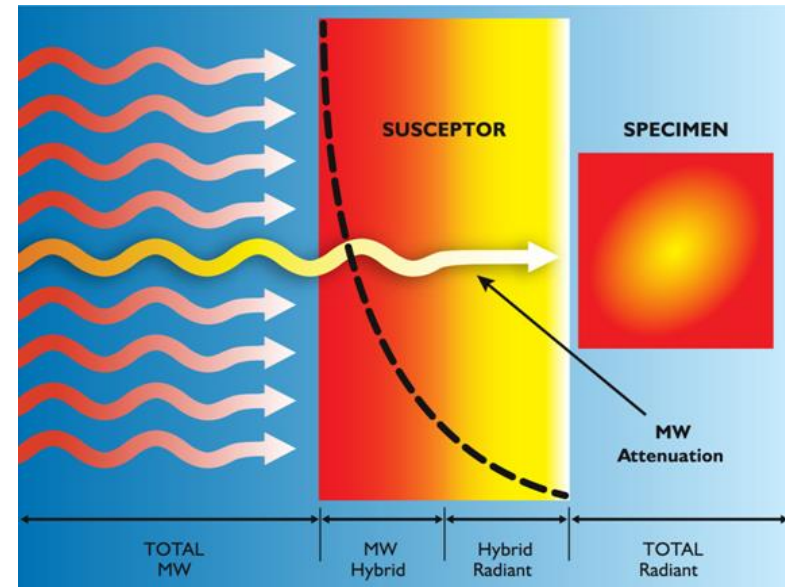
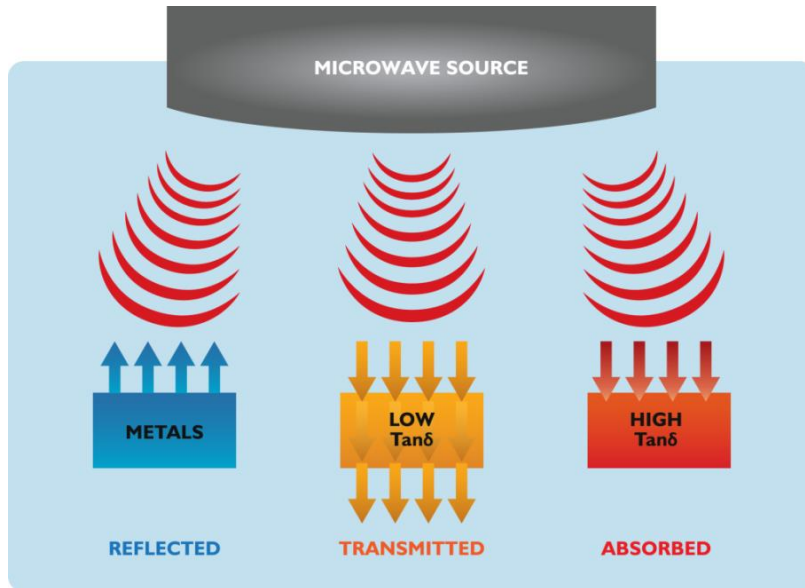




II. Proof of Concept (Continuous Flow Vertical Reactor)*

* US Patent Appln No: 14/624,092

MW Hybrid Technology – How it works?



$$P = 2\pi f[\epsilon_0 \epsilon_r' \tan\delta |E|^2 + \mu_0 \mu_r' \tan\phi [H]^2]$$

a simultaneous “EM + Heat” process

“Heat” → Microwave generated

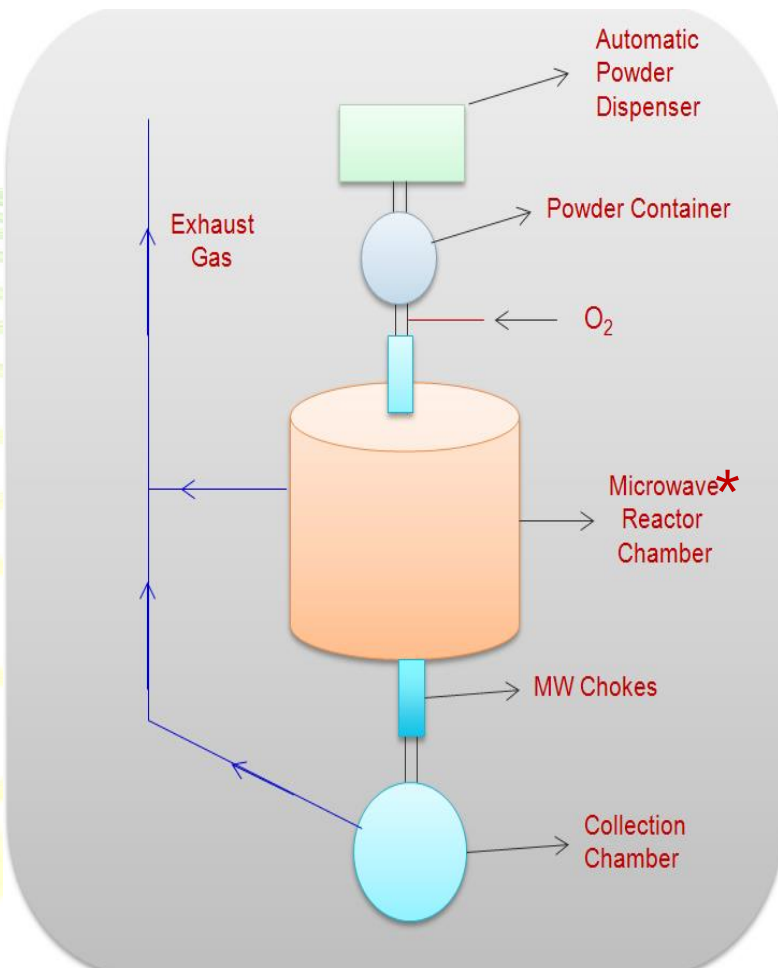
Reactant Materials



- Precursor powders from an external commercial source
- Intercalated with Sulfur

GRADE	AVERAGE PARTICLE SIZE (mm)	C/S RATIO	X -RATIO (cc/g)	NATURE
A	~ 300	31.61	300:1	BASIC
B	~ 300	28	300:1	ACIDIC
C	~ 300	27.14	255:1	BASIC
D	~ 200	25.80	225:1	BASIC
E	~ 100	27.58	100:1	ACIDIC

Graphene Exfoliation Vertical Reactor



* US Patent Appln. 14/047,172

Reactant



Product



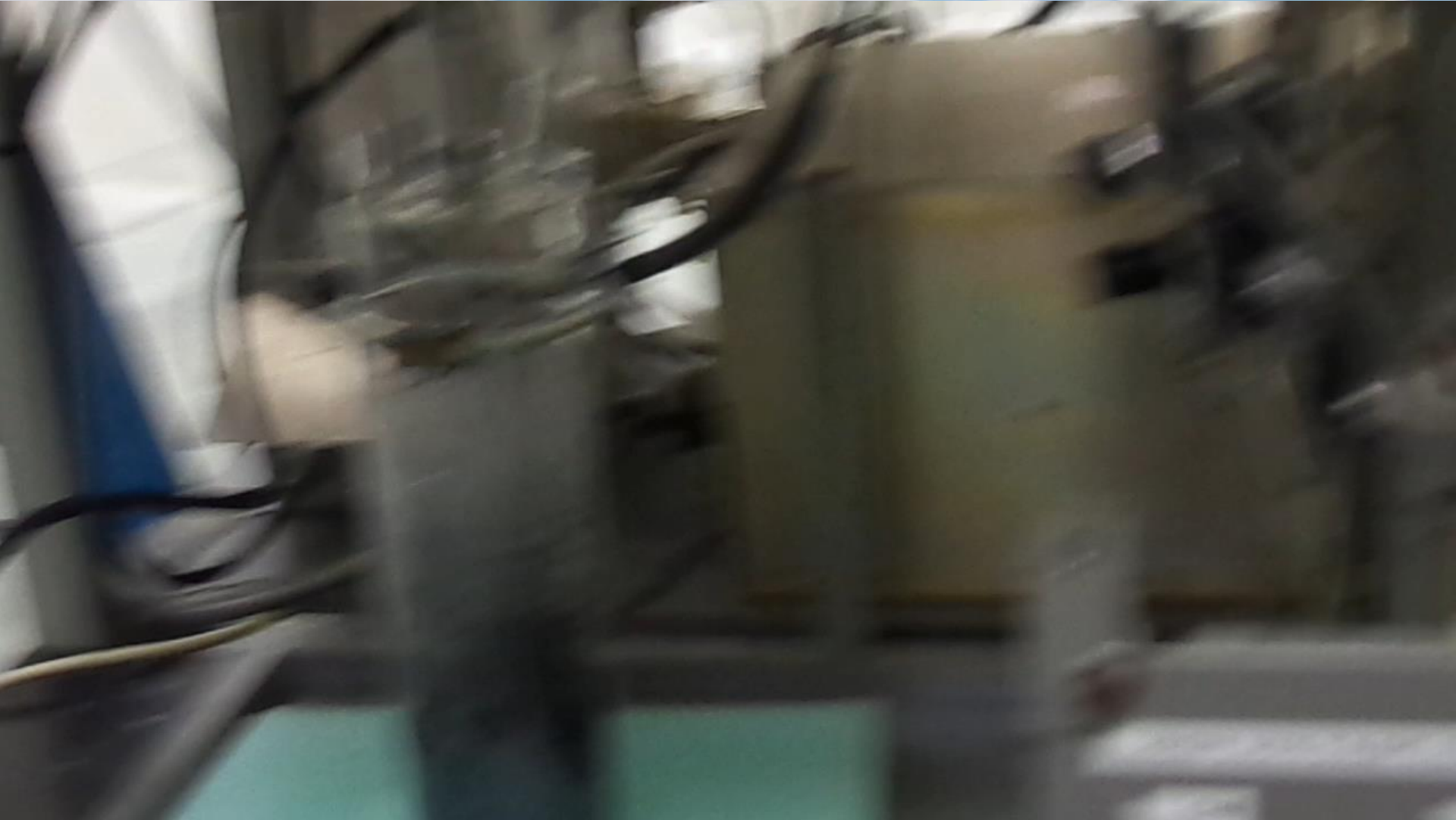
Variables

1. Particle Flow Rate
2. Inlet Gas Flow Rate
3. MW Power
4. Set Temp
5. Resonance Time

Exfoliation Characteristics

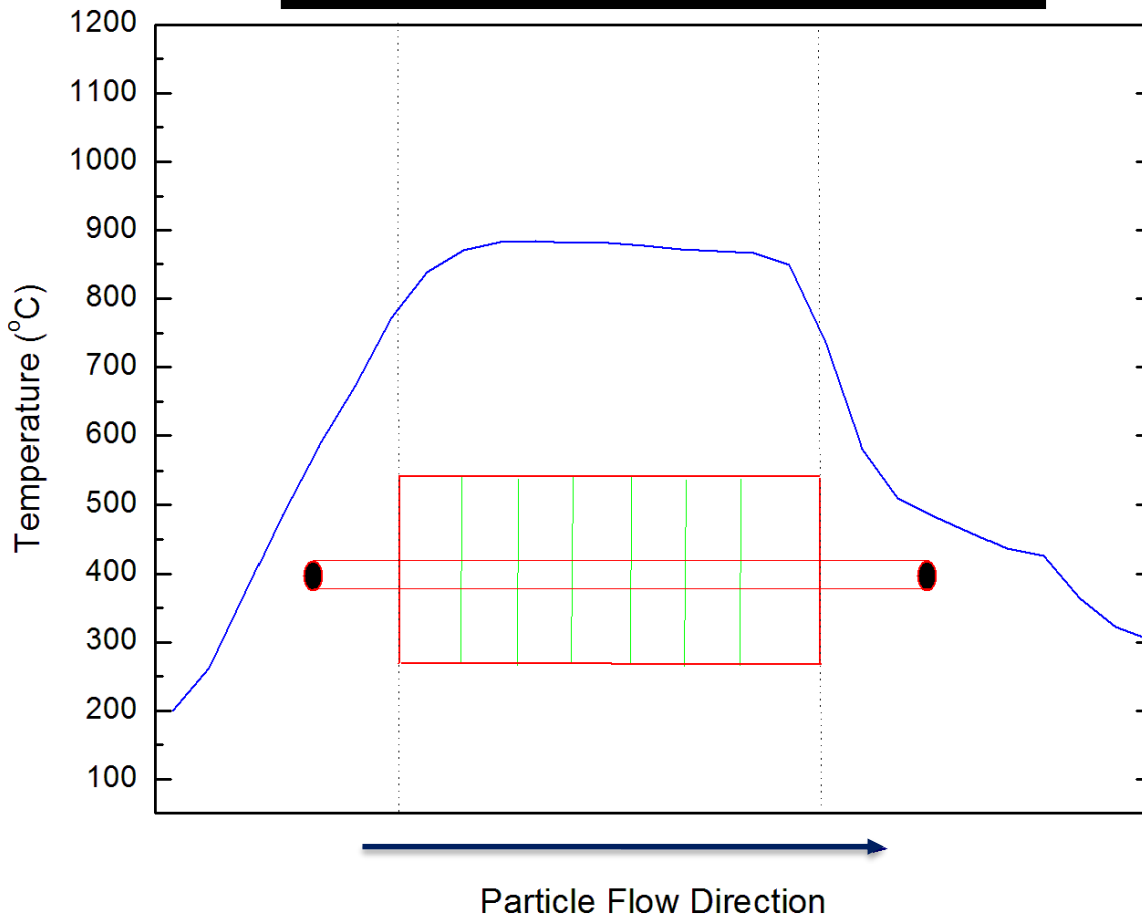
1. Instantaneous (<3s)
2. Continuous
3. Complete
4. MW effects
5. All types

Continuous Exfoliation



Exfoliation Reactor - Thermal Profile

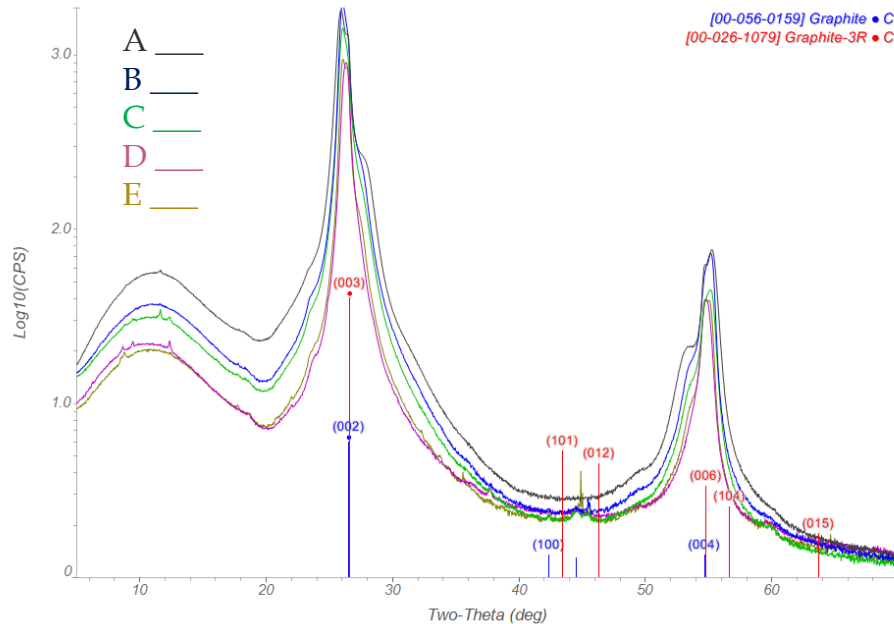
THERMAL PROFILE OF EXFOLIATION REACTOR



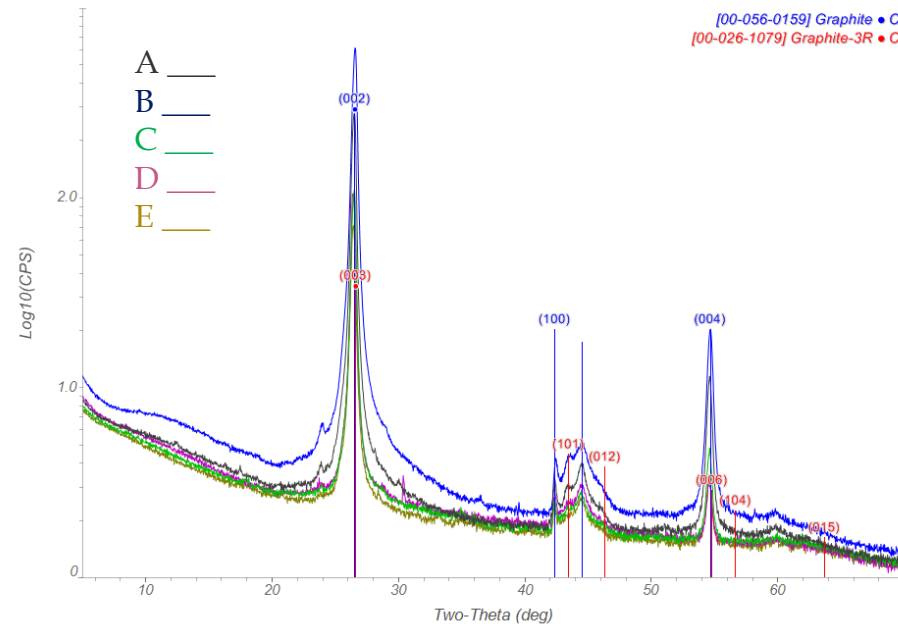
- “Hybrid Effect (EM + Thermal)”
- Temperatures measured using a thermocouple
- Controlled, uniform temperatures along material flow direction
- One shot exfoliation

Microwave Continuous Exfoliation : XRD

Reactants



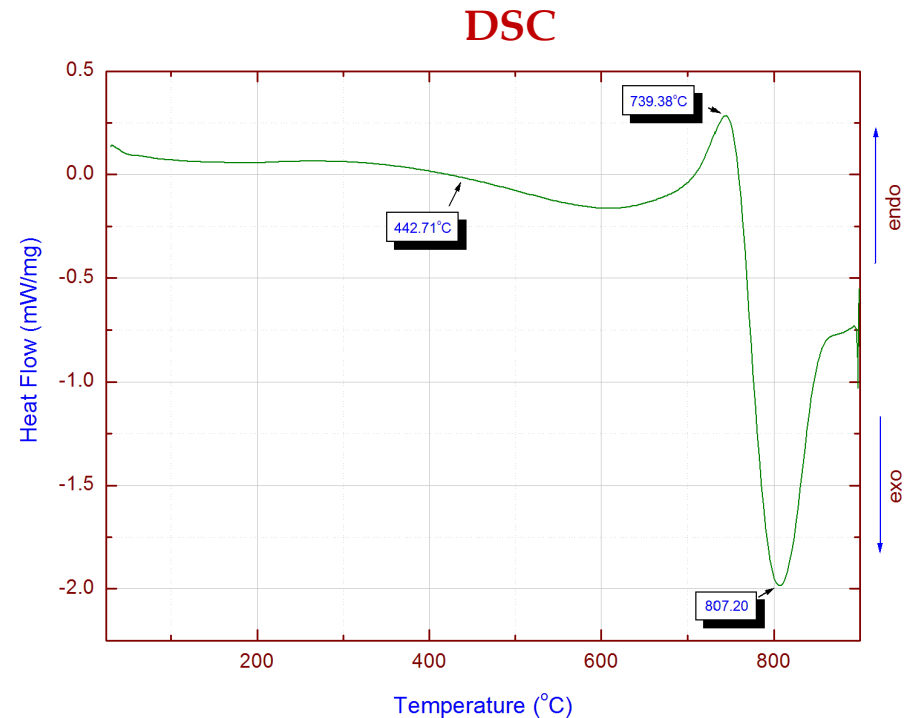
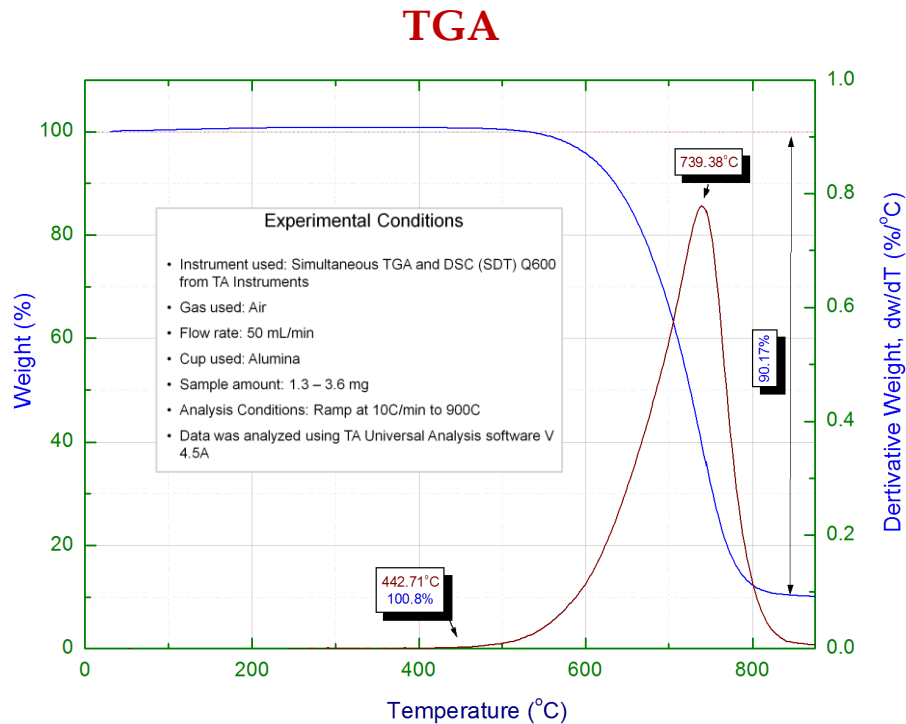
Products



- Emergence of new peaks; Opening up of new planes
- No additional phases seen
- Low angle broadening (amorphicity?)
- Change in lattice parameters
- FWHM (nano structures)

SAMPLE	X-RAY CRYSTALLITE SIZE (A°)
A	25.30
B	23.50
C	25.25
D	23.95
E	24.05

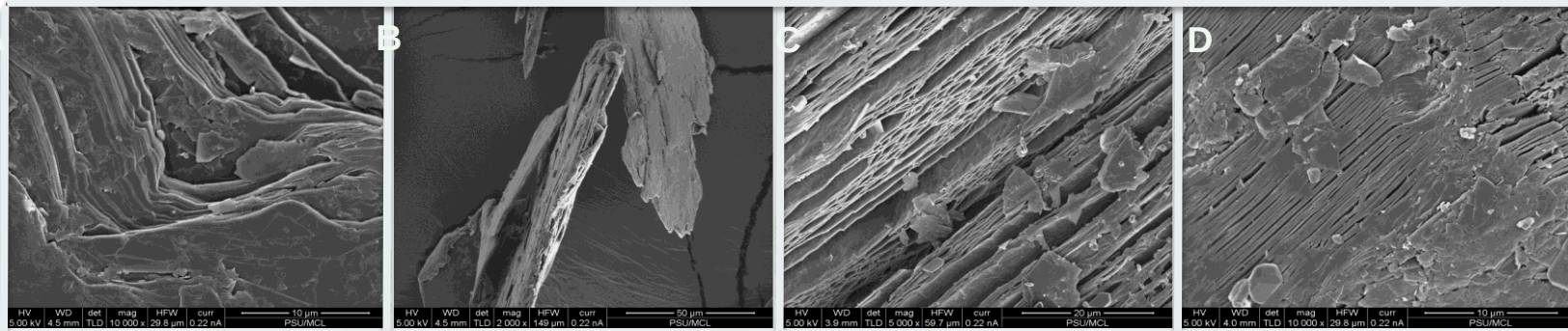
Microwave Continuous Exfoliation : TGA/DSC



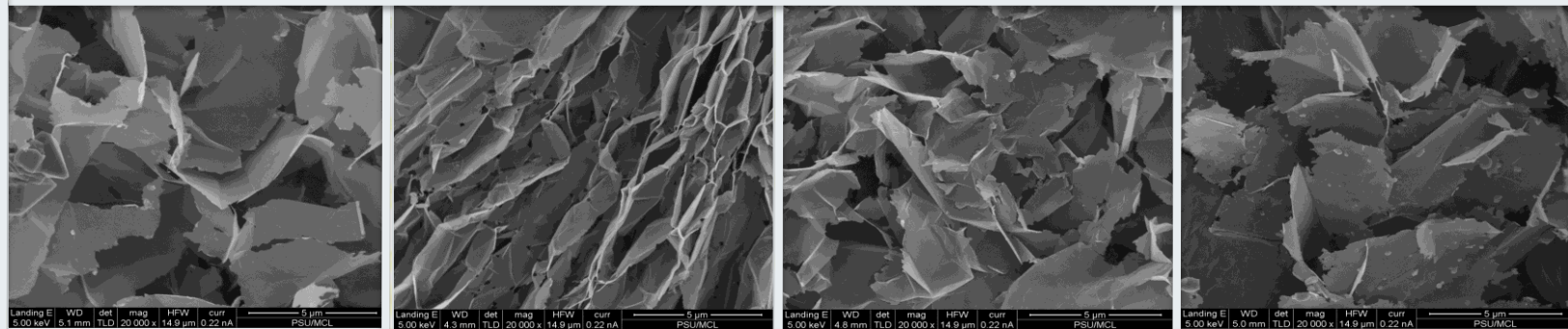
- Clean “S” trend; No heterogeneous phases present
- Weight loss starts at 442°C; Inflection Point 739°C
- ~ 10% residue
- Both “endo” and “exo” reactions observed

Microwave Continuous Exfoliation : SEM

Reactants



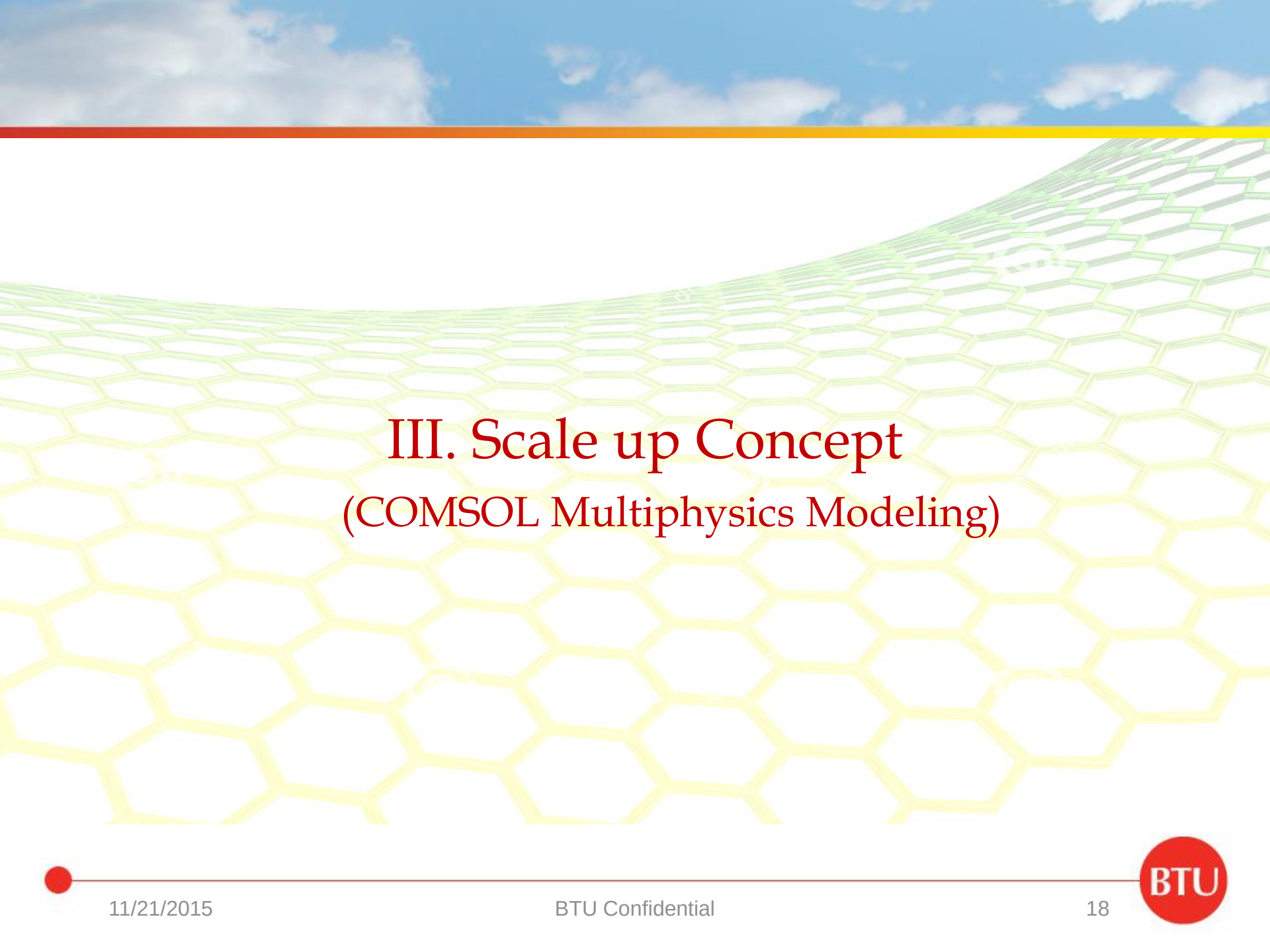
Products



- Reactant is a **layered graphitic structure** with hundreds of layers tightly stacked
- SEM reveals that each material has its own **unique** cross-sectional morphology (honeycomb)
- Before exfoliation, each layer is measured approximately **few microns** thick
- MW exfoliation process disintegrates the microstructure to form **fine platelet** structures (~5 μ m)
- These Graphene nanoplatelets are of high quality – **no secondary phases** detected
- Each exfoliated material has a **unique microstructure** deriving from the reactant material

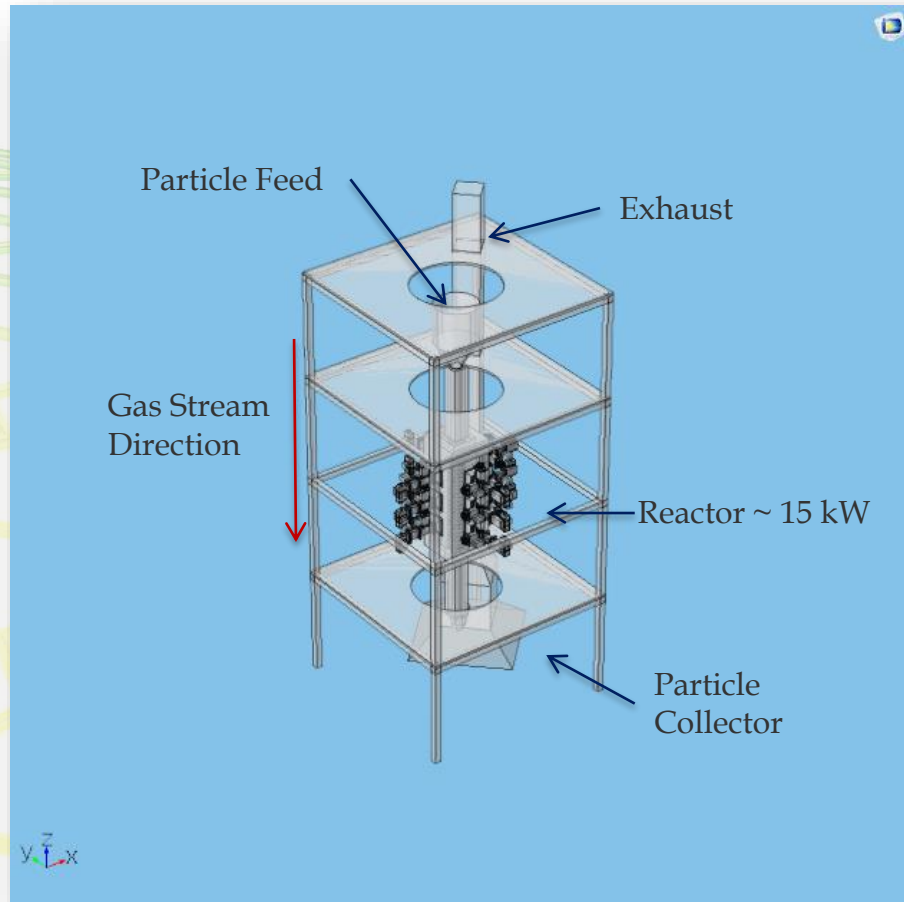
Summary of POC study

- Microwave continuous flow reactor for Graphene exfoliation
- Exfoliation of commercial grade reactants
- Exposure of newer planes
- No residual material left
- Alteration of interplanar distances
- Crystallite size $\sim 24 \text{ \AA}$
- “exo” and “endo” thermic reactions
- Dissociation of stacked graphitic layers
- Formation of “Nanoplatelets”
- Unique microstructures
- No detection of any secondary phases
- XRD/TGA/SEM results concur each other



III. Scale up Concept (COMSOL Multiphysics Modeling)

Flow Reactor – 60 Tons Per Year Capacity



- Typical Dimensions
 - Height ~ 16 Ft
 - Width ~ 6 Ft
 - Depth ~ 6 Ft
- Requirements
 - Power ~ 25,000 kW-Hr Annually
 - Recirculating Cooling Water
 - Gas Flow – Low Humidity Air
- Particle Process Time < 3 Sec
- Material Feed Requirement
 - Low Moisture Content
 - Feed Rate 2 g/s
- Particle Separation by mechanically scrubbing gas stream

Comsol Model: Flow Reactor Design

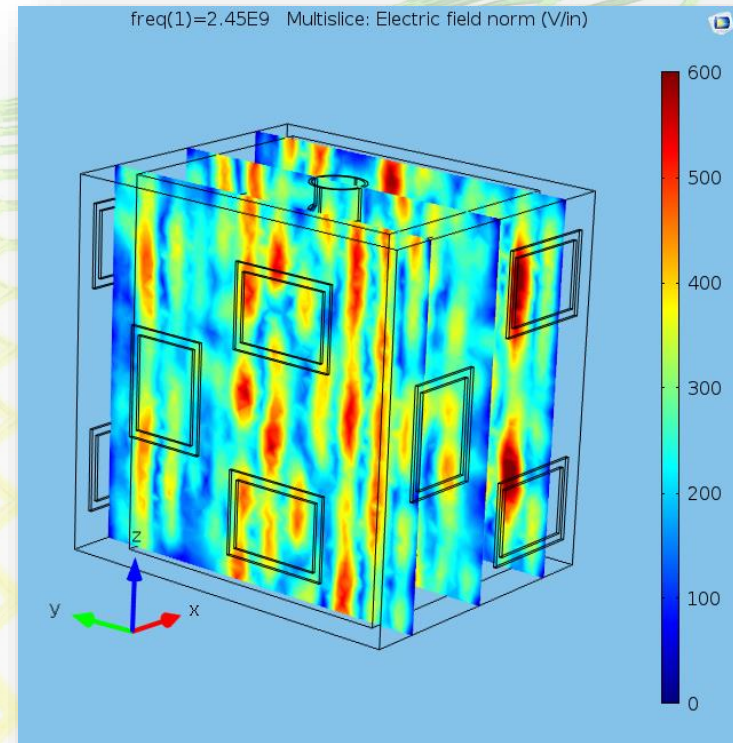
- Comsol Model Physics includes;

- Electro Magnetics
- Surface to Surface Radiation and Conductive Heat Transfer
- Fluid Flow with Particles

- Study Includes;

- Cavity Electric Field and Thermal Distribution
- Particle Dwell Time in Reactor
- Particle Heating

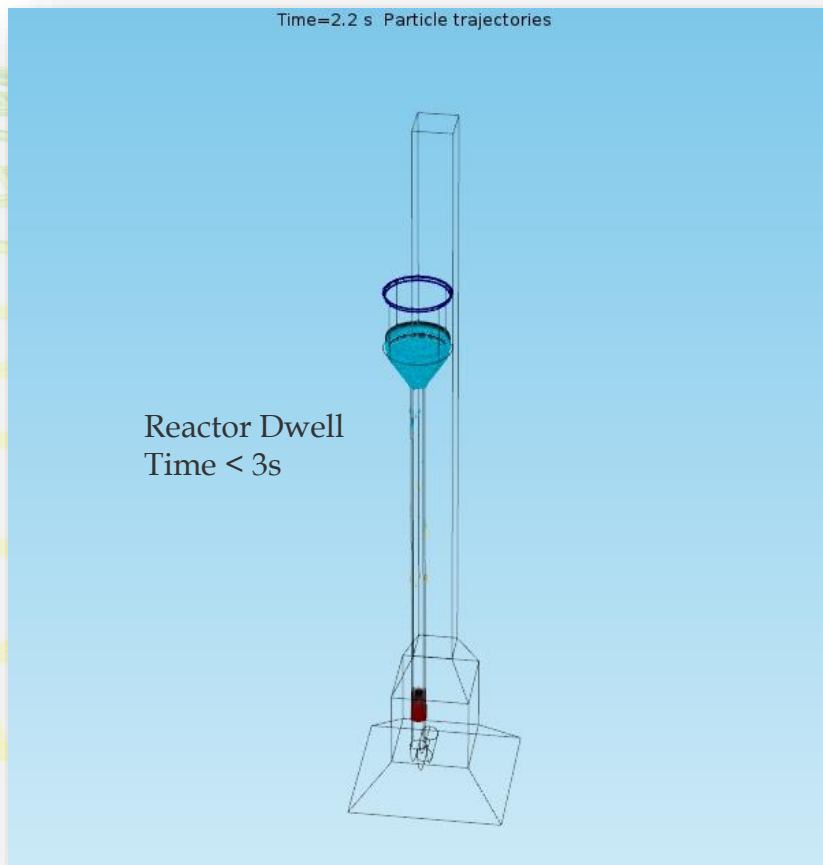
Electric Field Distribution



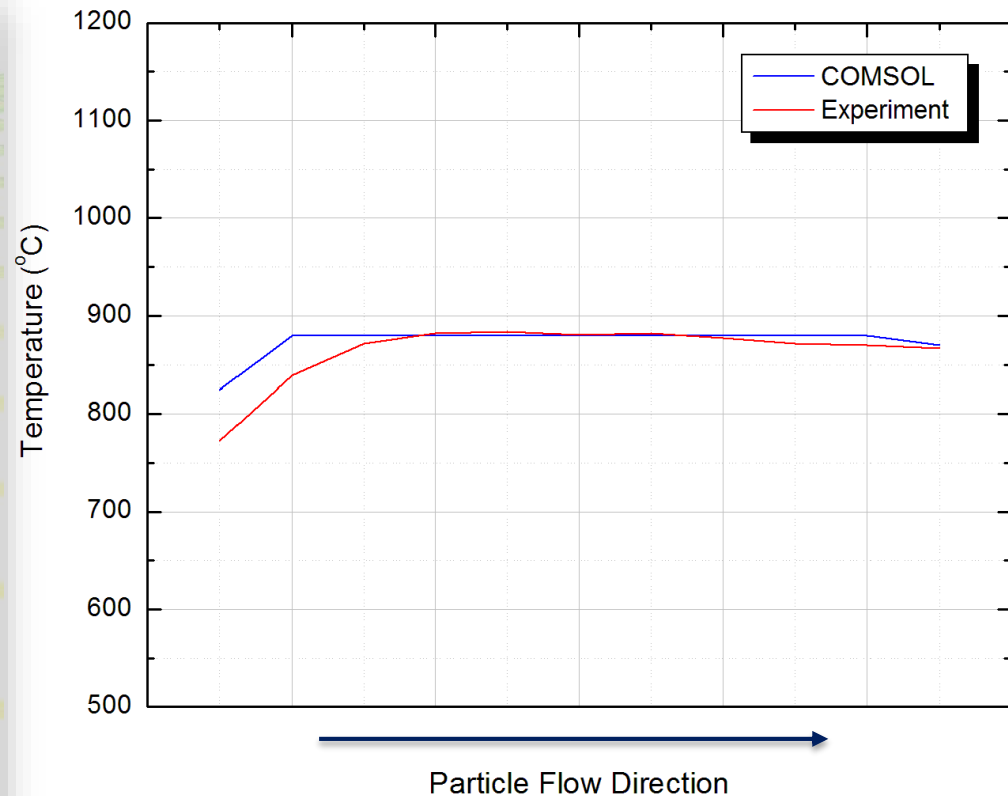
- Uniform Electric Field
- Simulation of Static Condition

Design by Analysis Results(Cont.)

Particle Flow Thru Reactor



Temperature Distribution



Experimental and simulated values match good

Flow Reactor Scaling Conditions

- 60 and 120 Tons/Year Systems Available
- Potential Scaling to 1000 Tons/Year
- Low Energy Requirements < \$2,500 / month (60 T)
- Equipment Acquisition Cost: \$11/kg / A. cap.
Target Cost: \$ 9/kg / A. cap. } + Licensing Fee



IV. Summary

Summary

- Concept Proven
- High Quality Graphene Nanoplatelets
- Continuous, Low Cost Process
- Scalability

For more information visit us at:



BOOTH No.

T35

- Production orders
- Partnerships

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Thank You!